

Slope-intercept form



1 b stands for y -intercept and m stands for slope.

2

a Slope = $-\frac{4}{3}$, y -intercept = -7

b Slope = 6 , y -intercept = 9

c Slope = -1 , y -intercept = 0

d Slope = -1 , y -intercept = -8

3

a
i $y = 15x - 10$

ii Slope = 15 , y -intercept = -10

b
i $y = 4x + 3$

ii Slope = 4 , y -intercept = 3

4

a $-\frac{1}{9}$

b $-\frac{1}{4}$

5

a $y = -8x - 9$

b $y = -5x + 7$

c $y = \frac{2x}{3} + 2$

d $y = -6x - 7$

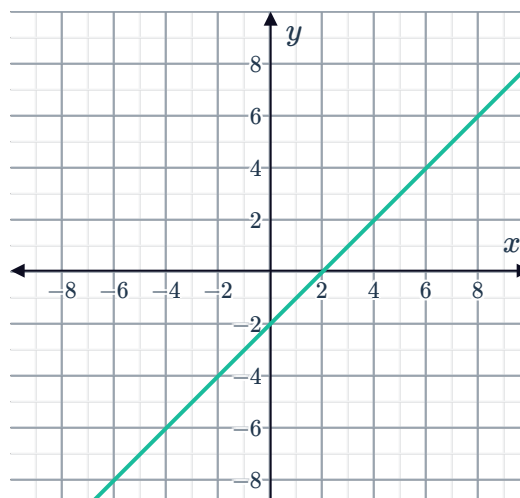
e $y = \frac{2}{13}$

6

a 9

b $y = 9x - 1$

7





8

a

i $\frac{2}{5}$

ii 2

iii $y = \frac{2x}{5} + 2$

b

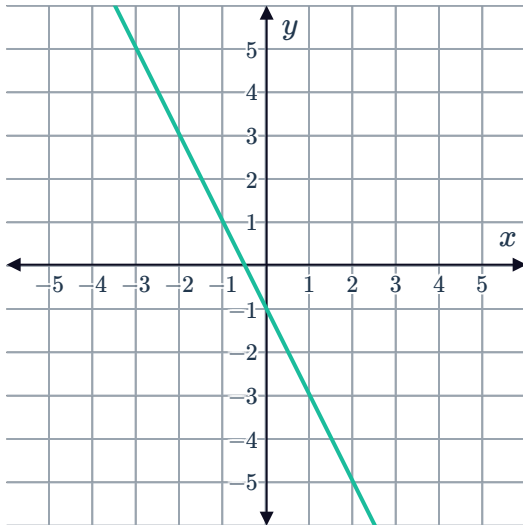
i 3

ii -1

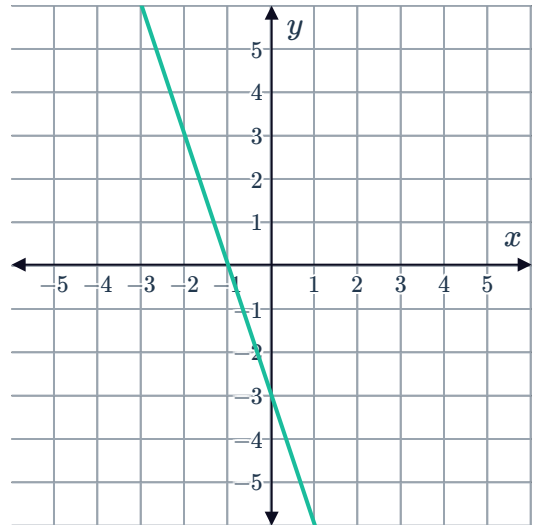
iii $y = 3x - 1$

9

a



b



10

a $b = -23$

b $y = -3x - 23$

11

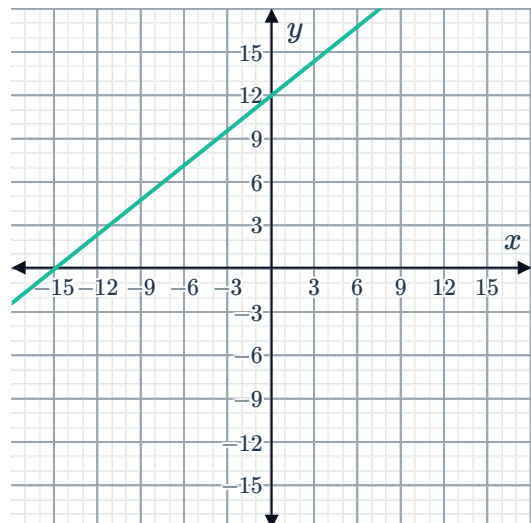
a

i $b = 12$

iii $(-15, 0)$

ii $y = \frac{4x}{5} + 12$

iv

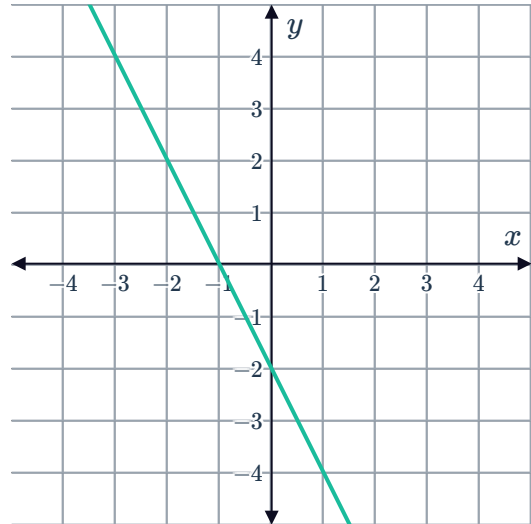


- b**
- i $b = -2$
 - iii $(-1, 0)$



ii $y = -2x - 2$

iv



12 a $y = -x + 2$

b $a = 8$

13 a $y = 1 + 8x$

b

x	0	1	2
y	-3	2	7

14 $y + 6x = 2$

15

x	0	1	2
y	3	7	11

16 Larger

17 Smaller

18 More rapidly

19 A

20 Smaller



21 Larger

22

- a 60 km/h
- c Bus

b 71 km/h

23

- a $u = 20$

b Compunet

24

- a 4 mg/min
- c In capsule form

b 5.9 mg/min

Additional questions

25 Line 1: $y = 2 + 6x$

Line 3: $y = 6x$

Line 6: $y = 6x - 2$

26

- a Yes, (0, 2)
- b Yes, (0, 0)
- c Yes, (0, 1)
- d No

27

- a Yes
- b No
- c Yes
- d No

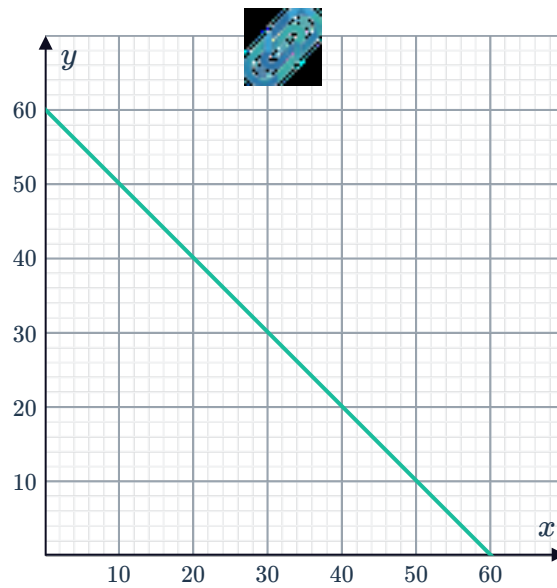
28

a

Number of minutes passed (x)	0	5	10	15	20	60
Amount of fuel left in tank (y)	60	55	50	45	40	0

- b $y = 60 - x$

c



29 $y = 20x + 110$

30

a $y = 30 - 5x$

- b The race is 30 miles and he is running at 5 miles per hour, so he will finish the race in 6 hours. Time cannot be negative and he will not be running any more after 6 hours, so the restriction is $0 \leq x \leq 6$.

31

a $y = -2.1x + 63$

- b The water level is falling at a constant rate of 2.1 feet per minute.
- c The water level was 63 feet before any water was pumped out.
- d 25.2 ft
- e An appropriate domain is $x \in [0, 30]$ because solving the equation $0 = -2.1x + 63$ we get $x = 30$ when the well is emptied. Once the well is emptied, the context is no longer valid. Time is always positive, so we get $[0, 30]$.

32

a $y = 35x + 250$

- b The total amount charged increases by \$35 for each additional hour of work.
- c The minimum amount charged by the carpenter, which is the initial fee of \$250.
- d The total cost is Materials+Labor. It will cost $\$1800 + 250 + 35 \times 12 = 2470$, and $2470 < 2500$, so she has budgeted enough for the ramp.

33



- a 1.63 ft/min
- b $y = 1.63x$
- c 9.78 ft
- d $x = 7$
- e The domain will be $x \in [0, 45]$ as after 45 minutes the diver will need to stop going deeper and start coming back to the surface.
- f $y = -1.63(x - 45) + 73.35 = -1.63x + 146.7$ for $x \in [45, 90]$

